Properties	Last Original Context Token	N = 0	N = 1	N = 2	N = 3	Examples
Lowercase No Space	12	14.5	18.1	13.1	13.4	‘itability’, ‘aka’, ‘ension’
Lowercase With Space	42	39.1	37.1	38.4	36.7	‘ sense’, ‘ tests’, ‘ punitive’
Uppercase No Space	2.4	2.7	2.2	2.8	1.6	‘V’, ’TABLE’, ’SE’
Uppercase With Space	1.9	2.4	1.1	1.5	1.7	‘ STAR’, ‘ UK’, ‘ USA’
Token length < 4	57.8	59.8	64.3	59.9	63.2	‘*’, ‘ate’, ‘</’
Token length ≥ 4	42.2	40.2	35.9	40.5	37	‘ validation’, ‘ Subaru’, ‘ulsion’
Punctuation	15.7	14.5	17.3	15.2	19	‘-’, ‘.’, ‘</’
Numerical	2.4	2.7	1.9	3.2	2.8	‘1998’, ‘001’, ‘5’

Table 1: Data Frequency of different token properties on the Last Prefix Tokens and GPT outputs at N=0,1,2,3. Each number in the table is a percentage of the test dataset, which is of size 1000.

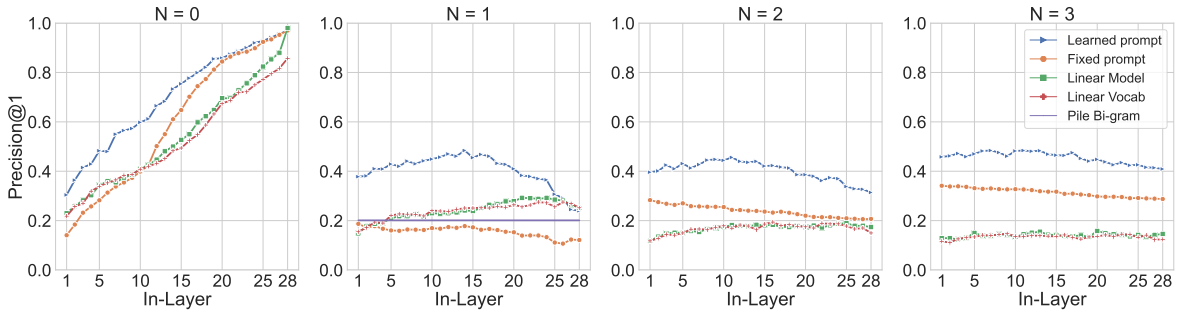

Figure 4: Accuracy (Precision@1) using the transplanted hidden representation. The $N = 0$ case models immediate next-token prediction, and $N \geq 1$ are the subsequent-token cases that are the focus of our work. The learned prompt is best able to recover future token information from hidden states of a preceding individual token, with predictive accuracy peaking at middle layers, with more than double the accuracy of a bigram baseline. A linear model predicting the hidden state fares comparably to predicting directly into the output vocabulary.

Learned Prompt Causal Intervention We then compare with trained prompts with the same token length as the fixed prompts. We train a soft prompt for each layer l from 1 to 28. Each learned prompt is trained by maximizing the probability of generating the token from the prefix context at the penultimate layer, when the hidden state is transplanted at layer l at the last token of the soft prompt, in the same way as the fixed prompts are applied. We train a prefix with a length of 10. This method performs best and is our main method.

3.4 Unveiling Subsequent Tokens

Figure 4 and Figure 5 illustrate the difference between our method and the baselines. The learned prompt optimized with the objective of predicting the next token (N=1) has the best performance. On average, the precision@1 is 24.8% higher, precision@5 is 25.3% higher, and precision@10 is 25.1% higher than the **best** baseline method. The bigram baseline[1] at N=1 is shown as a horizontal line; the bigram model i achieves 20.1% accuracy. For surprisal, the learned prompt also has the lowest value, which indicates its efficacy at maximally unveiling the information behind the hidden states.

[1]The bigram baseline is collected from 900,000 documents from the Pile dataset.

3.5 Contexts of Accurate Predictions

To further explore the contexts in which these methods seem better (or worse) able to predict subsequent tokens, we categorize input token (the last original context token) into eight (non-mutually exclusive) categories, shown in Table 3. We report the model accuracies when using layer 14, where the learned prompt model peaks.

While all categories of token types are predicted better by the learned prompt than by the linear model, the relative improvement is highest when the last context token is a lowercase token preceded by a space, or a longer token. This suggests that information about how to complete long words may not be immediately accessible by a linear model decoder, but that they can be made accessible by using the parameters of the pretrained model as done by the learned prompt intervention method.

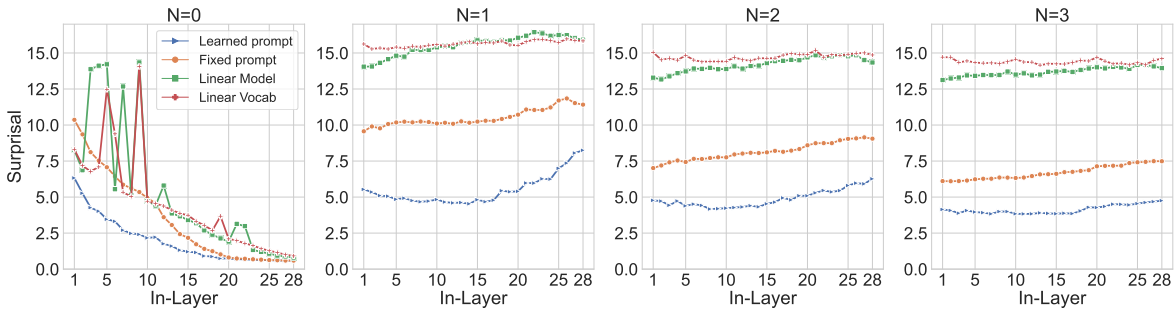

Figure 5: Average surprisal of the model after transplantation. Again the learned prompt performs best, confirming the presence of subsequent-token information encoded at middle-layer hidden states.

	LENS	N=1	N=2	N=3
Accuracy				
LEARNED	97.0	**48.4**	**43.7**	**46.9**
FIXED	97.0	20.8	30.0	36.5
HS	**98.0**	29.2	19.0	15.8
VOCAB	85.7	27.5	19.4	14.7
Surprisal				
LEARNED	**0.6**	**4.5**	**4.4**	**3.9**
FIXED	**0.6**	8.8	6.5	5.7
HS	0.8	14.1	13.2	13.1
VOCAB	0.9	15.3	14.4	14.2

Table 2: Best accuracy and surprisal results for each method. LEARNED refers to the Learned Prompt Causal Intervention Method; FIXED denotes the Fixed version. HS is the Linear Model variation that predicts Hidden State; VOCAB, is the Linear Model variation that predicts a distribution over the vocabulary directly.

We have also observed that the accuracy of predicting subsequent tokens is correlates with the model's confidence in its next token prediction. In the case of $N = 1$, for instance, the learned prompt intervention method's calibrated accuracy is 26%, 57%, 77%, and 95% for model confidence groups of 0-30%, 30-60%, 60-90%, and 90%-100%, respectively. These trends appear in $N = 2$ and $N = 3$ as well. This suggests that we might gainfully use this decoding method as a probing tool, trusting that predicted future tokens are generally accurate when the model is confident.

Does future information appear only in the presence of higher-level concepts? For example, one might hypothesize that in cases the language model predicts an entire named entity, that the probing method might decode future predictions more accurately. To investigate this, we performed sub-group analyses on test results to characterize how well the best probing method performed specifically for multi-token named entities. Interestingly, we found little difference: when examining just the named entity cases, we observe similar or slightly lower accuracy: 44%, 42% and 37% for $N = 1, 2, 3$, suggesting that future information is present broadly, not only for long entity names.

In sum, we have found that a single hidden state encodes information about outputs more than one token ahead, and we have demonstrated three different methods that can decode them for GPT-J-6B.

Application: Future Lens We apply Learned Prompt Intervention to create a novel probing tool we call the *Future Lens*. Given a soft prompt, we perform the intervention using the states arising from the user's prompt to provide a view into what the hidden states encode about future tokens. In Figure 6, we show an example for the prompt: "Marty McFly from". The Future lens reports the anticipated four tokens from every hidden state in the model (across layers).

In the Future Lens visualization, every cell represents a hidden state from a particular layer ("L{digit}") at a specific token. The shade of each cell indicates the model's average confidence with respect to the corresponding token predictions (darker shades indicate greater confidence). For example, at the cell representing the hidden state at Layer 25 at the token "from", we can see that the confidence in the predicted tokens "Back to the Future" is strong. This particular state suggests that the LLM already knows that Marty McFly is related to the Back to the Future movie. Interestingly, the model also assumes "Marty" to have the surname Donough. Returning to the predictions at token "from", we see that the early layers seem to first predict locations such as Australia or Boston. However, through future predictions, we can see the model begins to associate Marty McFly with a

Last Context Token Type	Linear: Vocab Space	Linear: Hidden State	Fixed Context	Learned Context
Lowercase No Space	21.7	25.2	9.2	**32.5**
Lowercase With Space	26.4	20.8	19.2	**51.9**
Uppercase No Space	**29.2**	26.3	0.0	**23.3**
Uppercase With Space	26.3	26.3	10.5	**31.6**
Token length < 4	26.5	24.9	21.8	**46.9**
Token length ≥ 4	23.9	24.4	18.0	**52.1**
Punctuation	28.7	28.7	16.6	**47.8**
Numerical	12.5	16.7	20.8	**33.3**

Table 3: Accuracy of predicting $N = 1$ token ahead (y_{T+1}, which predicts x_{T+2}) based on hidden representation of the last context token(x_T). Results are shown for layer $l = 14$, where the learned prompt model is most accurate.

movie around Layer 6. Hence, through this tool, we can gain further insights about the model's chain of predictions at every hidden state.

4 Related Work

Knowledge Prediction and Manipulation Recent works have delved into LLM internals to better understand how such models predict the next token at each computation step. Geva *et al.* (2021), for instance, find that the feed-forward layers in transformers operate as key-value memories, allowing one to intervene at those layers to modify the next token output (Geva et al., 2022). Frameworks such as ROME (Meng et al., 2022a) and MEMIT (Meng et al., 2022b) scale such manipulations to edit knowledge in stored in LLMs.

The consensus that has emerged in these papers is that some early-middle and late layer calculations contribute the most to the final predicted token. Tools such as Logit lens (nostalgebraist, 2020) and Tuned lens (Belrose et al., 2023; Din et al., 2023) allow us to look at the top-k values of the transformer at *every* layer and token to see early next-token predictions. Katz and Belinkov (2023) used logit lens to visualize semantic information flow in GPT-2 models. While these works primarily deal with next-token predictions, Hernandez et al. (2023) shows that specific attributes of an entity can be extracted with an affine transformation on the entity representation long before the LM is actually required to predict the attribute, enabling an "attribute lens" on early layers and early tokens. We aim to characterize how the current hidden state would affect the prediction of both the next token and tokens farther ahead, but unlike Hernandez et al. (2023), we deal with open contexts and are not constrained to certain relations.

Early Exit Decoding To optimize the running time and space requirements of training models, prior work has looked at "early exit" strategies, which usually involves stopping at earlier layers of computation and estimating the final predictions based on those computations (Schuster et al., 2022; Xin et al., 2021; Kong et al., 2022; Zhang and He, 2020; Din et al., 2023). The takeaway from these methods is that it is possible to achieve prediction performance comparable to that observed when all layers are used even when dropping a couple of computational layers for each token. For instance, Din and colleagues (2023) used linear transformations to predict a later layer's hidden representation from an earlier layer at the same token. This approach was able to preserve $\sim$95% of the full transformer model outputs on GPT-2 (Radford et al., 2019) and BERT (Devlin et al., 2018). This result implies that initial model layers encode information that to largely determines the final output. In this work we test the limits of this phenomenon by evaluating the degree to which a single hidden state for a token at position T can be used to predict tokens multiple steps ahead (i.e., at $T + N$).

Memorization in Language Models Due to the potentially sensitive information present in the datasets used to train language models (LMs), past work has investigated what, when, and why memorization occurs (Carlini et al., 2021, 2019; Feldman and Zhang, 2020; Lehman et al., 2021), how memorization changes as a function of training data size (Carlini et al., 2023; Wei et al., 2022), and how other memorized information can be detected based on model internal states (Haviv et al., 2023).

These works have collectively illustrated that there are some text snippets that LMs remember and can output verbatim or in closely paraphrased versions ("approximate memorization"; Ippolito

	Mart	y	Mc	Fly	from
L1	inez the court held	erson the screen.	Afee the same source	er and the other	behindrrhaph
L2	inez\n\nThe	\xe9n-1-	Afee and the other	er the left of	Sons\n\nThe
L3	.\n\nThe	Friedman and the other	Afee\n	er\n\n\n	Havanaa:\n
	\n de la c	Barn and the other	Lean.\n\n	er\ufffd\ufffd said	1992a and the
	\n de la c	ring and the other	Leanmig.	er \ufffd\ufffdI	Oklahoma first time I
	\n de la c	ring and the next	Afee\n\nThe	world and the future	Austria book.\n
	\n de la c	ring and the other	Leanaway from the	mer 1, 1	Australia movie.\n
	\n, and the	ell and the other	Lean\n\nThe	walker the time of	England first time he
	\n, and the	ellLean, and	Lean\n\nThe	walker be the first	Australia movie "The
	\n, and the	ellDonough,	Lean\n\nThe	te Marty McFly	Australia movie "The
	\n"\n	ellDonough,	Lean\n\nThe	te Marty McFly	Vietnam movie "The
	\n" id="	GreenbergDonough,	Bride\n\nThe	te Marty McFly	Germany movie "The
	\n" id="	GreenbergDonough,	Lean\n\nThe	movies Marty McFly	Boston movie "Back
	\n" id="	ellDonough,	Bride\n\nThe	movie Marty McFly	movie movie \ufffd\ufffd
	\n" id="	riumDonough,	Bride\n\nThe	movie Marty McFly	movie movie "The
	\n" id="	WalshDonough,	Bride\n\nThe	movie Marty McFly	movie movie "The
	\n" id="	McDonough,	Bride\n\nThe	movie Marty McFly	movie Back to the
	\n" id="	McDonough,	Bride\n\nThe	movie Marty McFly	movie movie "Back
	\n" id="	riumDonough,	Flylew\n	movie Marty McFly	1980 Back to the
	\npng" alt	ring Marty, and	Fly\n\nThe	movie Marty McFly	movie Back to the
	\npng" alt	Mc Marty, and	Fly\n\nThe	Returns Marty McFly	movie Back to the
	\npng" alt	Mc\ufffd\ufffd he	Fly\n\nThe	arrives Marty McFly	movie Back to the
	\npng" alt	Mc he was a	Fly\n\nThe	arrives Marty McFly	1984 Back to the
	\npng" alt	Mc and I'm	Fly\n\nThe	arrives Marty McFly	1989 to the Future
	\npng" alt	Mc and I'm	Fly\n\nThe	's\nThe first	Back to the Future
L26	\n1.0	Mc and I'm	Fly\n\nThe	(\nThe first	Back movie "Back
L27	.\n\nThe	Mc and the other	Fly.\n\n	,\nThe first	the movie "Back
L28	y\n\nThe	Mc and I'm	Fly.\n\n	\n and the future	Back future.\n

Figure 6: The Future Lens applied to the hidden states of GPT-J-6B processing *Marty McFly from*. Each cell illustrates the most likely sequence of future tokens that the respective hidden state predicts. The darker boxes correspond to higher probabilities/confidence.

et al. 2023). Other work (Haviv et al., 2023) has shown that earlier layers of models tend to promote memorized concepts or tokens, while later layers boost model confidence in these tokens. Our paper can be viewed as an extension of this work on investigating memorization of multi-token phrases: we ask whether and to what extent a single model hidden state encodes multi-token information.

Prompt Tuning Prompt Tuning has emerged as a parameter-efficient method for fitting LMs for new downstream tasks. By freezing the LM and optimizing only the soft prompt parameters, models are able to achieve performance comparableto that observed after fine-tuning all parameters. Li *et al.* (2021) introduced prefix tuning which entailed training plug-and-play prefix that steers the behavior of the LMs for the downstream tasks. Other work (Wallace et al., 2019) applied a gradient-based method to search for the best discrete prompts which enable the model to produce desire generation. Sun and colleagues (2023) train the prefix soft prompt as a way of aligning semantically equivalent instructions in latent space.

5 Discussion

In this paper, we explored the degree to which we are able to decode multi-token outputs subsequent to a particular token on the basis of its hidden representation alone. The results in Table 2 and Figures 4 and 5 indicate that the representations encode such information, to some degree. Among the decoding methods we assessed, learned prompts are best able to predict such future tokens. Both the linear and the learned prompt models achieve better accuracy than the empirical bigram baseline at $N = 1$. Interestingly, predictive accuracy of the learned prompt model peaks at the middle-layer hidden states, suggesting that subsequent-token information is encoded at those layers; this pattern is very different from the immediate next-token $N = 0$, in which accuracy peaks at the last layer.

The learned prompt model realizes an accuracy sufficient to be potentially useful as a "Future Lens" to provide insights about subsequent token information contained in hidden states within LLMs. This provides a way to decode a short sequence of tokens encoded in a hidden state, rather than only the single immediate token prediction.

Data and Code Availability All code and data for demo and implementation is made available at: https://future.baulab.info

Acknowledgements This work was supported by Open Philanthropy and by the National Science Foundation (NSF) award 1901117. We thank the Center for AI Safety (CAIS) for making computing capacity available for this research.

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